LESSON 3.10

Convert Measurement Units

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can convert measurement units using ratios and conversion factors.

WHY THIS MATTERS You're building a brand-new arcade cabinet! The blueprints use feet, but the parts catalog lists measurements in inches. The wiring instructions use yards, and the weight limits are in pounds — but the shipping labels use ounces. You need to convert between units to make sure everything fits and ships correctly. The blueprint says the cabinet is 6 feet tall, and the parts catalog lists a side panel that is 70 inches tall. Does the panel fit?

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Conversion Factor *Spanish: Factor de conversión*	A ratio that helps you change one unit into another.	1 foot = 12 inches, so the factor is 12
Unit *Spanish: Unidad*	A standard amount used to measure, like an inch or a liter.	Inches, feet, pounds, ounces
Equivalent *Spanish: Equivalente*	Having the same value, just in different units.	2 feet = 24 inches
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	12 inches per 1 foot
Customary units *Spanish: Unidades usuales (sistema inglés)*	The U.S. way of measuring, like inches, feet, and pounds.	12 inches = 1 foot, 3 feet = 1 yard, 16 ounces = 1 pound
Metric units *Spanish: Unidades métricas*	A way of measuring by 10s, like meters, liters, and grams.	1 meter = 100 centimeters, 1 kilogram = 1,000 grams

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

The cabinet is 5 feet tall. I want to know how many inches that is.

1. My conversion factor is 1 foot = 12 inches.
2. Inches are smaller than feet, so I multiply: 5 × 12 = 60.
3. So 5 feet = 60 inches.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A wire is 4 yards long. The factor is 1 yard = 3 feet.

1. _____
2. _____
3. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- When converting from a _____ unit to a _____ unit, I _____ by the conversion factor because _____.

Word bank Conversion Factor | Unit | Equivalent | Ratio | Customary units | Metric units | larger | smaller | multiply | divide | conversion factor | ratio

WATCH OUT!

A common mistake

A common mistake in Convert Measurement Units is applying the conversion factor in the wrong direction

LESSON 3.10

Convert Measurement Units

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Complete the table. Show how you found each value.**

Complete the conversion table using the given conversion factors.

Given	Conversion Factor	Answer
2 feet	1 ft = 12 in	24 inches
6 yards	1 yd = 3 ft	
3 pounds	1 lb = 16 oz	

HINT Each row hands you the conversion factor — the finished row shows 2 feet × 12 = 24 inches.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

How many inches are in 7 feet?

(A) 70 inches
(B) 72 inches
(C) 84 inches
(D) 96 inches

HINT The key fact: 1 foot = 12 inches. You have 7 feet and want inches.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

How many feet are in 3 yards?

(A) 6 feet
(B) 9 feet
(C) 12 feet
(D) 36 feet

HINT The key fact: 1 yard = 3 feet. You have 3 yards and want feet.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

How many ounces are in 4 pounds?

(A) 16 ounces
(B) 32 ounces
(C) 48 ounces
(D) 64 ounces

HINT The key fact: 1 pound = 16 ounces. You have 4 pounds and want ounces.

SHOW YOUR WORK

5 Circle the letter of the best answer. Show how you know.

Convert 24 inches to feet.

(A) 2 feet
(B) 3 feet
(C) 12 feet
(D) 24 feet

HINT This one goes the OTHER direction: inches (small) to feet (big), so the number of units gets smaller.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Common Mistake

1. **Convert 2 yards of rope to feet** 2 yards → feet
2. **Use the conversion factor** 1 yard = 3 feet
3. **Operate** 2 ÷ 3 = 0.67 feet
4. **Final answer** 2 yards ≈ 0.67 feet

Which step has the mistake? Explain what went wrong, then show the correct work.

HINT You are going from a bigger unit (yards) to a smaller unit (feet). Should the number get bigger or smaller?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

SENTENCE STARTER When converting from a ___ unit to a ___ unit, I ___ by the conversion factor because ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.10

Convert Measurement Units

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the letter of the matching item on each line.**

Match each measurement to its equivalent value.

_____ 3 yards	(A) 9 feet
_____ 4 feet	(B) 48 inches
_____ 5 pounds	(C) 80 ounces
_____ 2 yards	(D) 72 inches
_____ 36 inches	(E) 3 feet
_____ 3 pounds	(F) 48 ounces

2 **Complete the table. Show how you found each value.**

An arcade cabinet has parts measured in different units. Convert each to the target unit.

Part	Measurement	Target Unit	Converted
Power cord	2 yards	inches	
Top panel	54 inches	feet	
Weight	96 ounces	pounds	
Side rail	5 feet	inches	

SHOW YOUR WORK

3 **Write the name of the correct group on each line.**

When converting, do you multiply or divide? Sort each conversion by the correct operation.

Groups: Multiply | Divide

_____ Feet → inches

_____ Yards → feet

_____ Pounds → ounces

_____ Inches → feet

_____ Ounces → pounds

_____ Feet → yards

PART 2 Apply it

real-world problems

4 **Circle the letter of the best answer. Show how you know.**

A shelf is 30 inches long. How many feet and inches is that?

(A) 1 foot 18 inches

(B) 2 feet 6 inches

(C) 2 feet 8 inches

(D) 3 feet 0 inches

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Common Mistake

1. **Convert 80 ounces of bolts to pounds** 80 ounces → pounds
2. **Use the conversion factor** 1 pound = 16 ounces
3. **Operate** 80 × 16 = 1,280 pounds
4. **Final answer** 80 ounces = 1,280 pounds

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

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LESSON 3.10

Convert Measurement Units

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
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PART 1 Understand the idea

models and meaning

1 **Mark and label each value on the number line.**

Place these lengths on a number line in inches: A = 1 foot, B = 1.5 feet, C = 2 feet, D = 1 yard



SHOW YOUR THINKING

PART 2 Explain and correct

reasoning

2 Find the mistake. Explain it, then show the correct work.

Find the Conversion Error

1. **Convert 6 feet to yards** 6 feet → yards
2. **Use conversion factor** 1 yard = 3 feet
3. **Multiply** 6 × 3 = 18 yards
4. **Final answer** 6 feet = 18 yards

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 Answer in complete sentences.

A recipe says you need 5 feet of ribbon, but the store only sells ribbon in inches. Another recipe says you need 2 yards of fabric, but your ruler measures in feet. Convert both measurements and explain the steps you used. Which conversion factor did you use each time?

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

How do you decide whether to multiply or divide when converting units?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help